

SOLIDWORKS API (.NET VBA C#) — Reference (15/337): IAssemblyDoc (part 3 of 5)

SOLIDWORKS API Documentation (converted from CHM)

IAssemblyDoc (part 3 of 5) (Type)

SOLIDWORKS API Help

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GetExplodedViewCount Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetExplodedViewCount Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::GetExplodedViewCount2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetExplodedViewCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim value As System.Integer

value = instance.GetExplodedViewCount()
```

C#

```
System.int GetExplodedViewCount()
```

C++/CLI

```
System.int GetExplodedViewCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of exploded views

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetExplodedViewCount](#).

Example

[Create Exploded Views of an Assembly \(VBA\)](#)
[Create Exploded Views of an Assembly \(VB.NET\)](#)
[Create Exploded Views of an Assembly \(C#\)](#)

Remarks

To take advantage of the explode mechanism, see [IAssemblyDoc::CreateExplodedView](#), [IAssemblyDoc::GetExplodedViewNames](#), [IAssemblyDoc::IGetExplodedViewNames](#), [IAssemblyDoc::AutoExplode](#), [IAssemblyDoc::ShowExploded2](#), [IAssemblyDoc::ViewExplodeAssembly](#), [IAssemblyDoc::ViewCollapseAssembly](#), [IView::ShowExploded](#), [IView::IsExploded](#), and [IModelDoc2::IsExploded](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

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GetExplodedViewCount2 Method (IAssemblyDoc)

[See Also](#) [Example](#)



Gets the number of exploded views in the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function GetExplodedViewCount2( _  
    ByVal ConfigurationName As System.String _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim ConfigurationName As System.String  
Dim value As System.Integer  
  
value = instance.GetExplodedViewCount2(ConfigurationName)
```

C#

```
System.int GetExplodedViewCount2(  
    System.string ConfigurationName  
)
```

C++/CLI

```
System.int GetExplodedViewCount2(  
& System.String^ ConfigurationName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigurationName
Name of the configuration

Return Value

Number of exploded views in the specified configuration

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetExplodedViewCount2](#).

Example

[Get Exploded Views for Configuration \(C#\)](#)
[Get Exploded Views for Configuration \(VB.NET\)](#)
[Get Exploded Views for Configuration \(VBA\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::GetExplodedViewNames2 Method \(\)](#)
[IAssemblyDoc::GetExplodedViewConfigurationName Method \(\)](#)
[IAssemblyDoc::CreateExplodedView Method \(\)](#)
[IAssemblyDoc::AutoExplode Method \(\)](#)
[IAssemblyDoc::ShowExploded2 Method \(\)](#)
[IAssemblyDoc::ViewCollapseAssembly Method \(\)](#)
[IAssemblyDoc::ViewExplodeAssembly Method \(\)](#)
[IModelDocExtension::IsExploded Method \(\)](#)
[IView::IsExploded Method \(\)](#)
[IView::ShowExploded Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetExplodedViewNames Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::GetExplodedViewNames2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetExplodedViewNames() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim value As System.Object
```

```
value = instance.GetExplodedViewNames()
```

C#

```
System.Object GetExplodedViewNames()
```

C++/CLI

```
System.Object^ GetExplodedViewNames();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of strings of the names of the exploded views in the active configuration

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetExplodedViewNames](#).

Example

[Create Exploded Views of an Assembly \(VBA\)](#)
[Create Exploded Views of an Assembly \(VB.NET\)](#)
[Create Exploded Views of an Assembly \(C#\)](#)

Remarks

To take advantage of the explode mechanism, see [IAssemblyDoc::CreateExplodedView](#), [IAssemblyDoc::IGetExplodedViewNames](#), [IAssemblyDoc::GetExplodedViewCount](#), [IAssemblyDoc::AutoExplode](#), [IAssemblyDoc::ShowExploded2](#), [IAssemblyDoc::ViewExplodeAssembly](#), [IAssemblyDoc::ViewCollapseAssembly](#), [IView::ShowExploded](#), [IView::IsExploded](#), and [IModelDoc2::IsExploded](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

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GetExplodedViewNames2 Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetExplodedViewNames2 Method (IAssemblyDoc)

Gets the names of the exploded views in the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function GetExplodedViewNames2( _  
    ByVal ConfigurationName As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim ConfigurationName As System.String  
Dim value As System.Object  
  
value = instance.GetExplodedViewNames2(ConfigurationName)
```

C#

```
System.object GetExplodedViewNames2(  
    System.string ConfigurationName  
)
```

C++/CLI

```
System.Object^ GetExplodedViewNames2(  
& System.String^ ConfigurationName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigurationName
Name of the configuration

Return Value

Array of strings of the names of the exploded views in the specified configuration

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetExplodedViewNames2](#).

Example

[Get Exploded Views for Configuration \(C#\)](#)
[Get Exploded Views for Configuration \(VB.NET\)](#)
[Get Exploded Views for Configuration \(VBA\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::AutoExplode Method \(\)](#)
[IAssemblyDoc::CreateExplodedView Method \(\)](#)
[IAssemblyDoc::GetExplodedViewConfigurationName Method \(\)](#)
[IAssemblyDoc::GetExplodedViewCount2 Method \(\)](#)
[IAssemblyDoc::ShowExploded2 Method \(\)](#)
[IAssemblyDoc::ViewCollapseAssembly Method \(\)](#)
[IAssemblyDoc::ViewExplodeAssembly Method \(\)](#)
[IModelDocExtension::IsExploded Method \(\)](#)
[IView::IsExploded Method \(\)](#)
[IView::ShowExploded Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

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GetFeatureScope Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetFeatureScope Method (IAssemblyDoc)

Gets the components affected by this feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFeatureScope( _  
    ByVal FeatureIn As Feature _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim FeatureIn As Feature  
Dim value As System.Object  
  
value = instance.GetFeatureScope(FeatureIn)
```

C#

```
System.Object GetFeatureScope(  
    Feature FeatureIn  
)
```

C++/CLI

```
System.Object^ GetFeatureScope(  
& Feature^ FeatureIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatureIn
[IFeature](#)

Return Value

Array of [components](#) affected by this feature

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetFeatureScope](#).

Remarks

Feature scope information is only provided if the assembly is opened in its own window. This method gets both the components affected and potentially affected by this feature. Thus, not all of the components in this list are necessarily affected by this feature.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::GetFeatureScope Method](#)
[IAssemblyDoc::AddToFeatureScope Method](#)
[IAssemblyDoc::GetFeatureScopeCount Method](#)
[IAssemblyDoc::UpdateFeatureScope Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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GetFeatureScopeCount Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetFeatureScopeCount Method (IAssemblyDoc)

Gets the number of components affected by this feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFeatureScopeCount( _  
    ByVal FeatureIn As Feature _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim FeatureIn As Feature  
Dim value As System.Integer  
  
value = instance.GetFeatureScopeCount(FeatureIn)
```

C#

```
System.int GetFeatureScopeCount(  
    Feature FeatureIn  
)
```

C++/CLI

```
System.int GetFeatureScopeCount(  
& Feature^ FeatureIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatureIn
[IFeature](#)

Return Value

Number of components affected by this feature

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetFeatureScopeCount](#).

Remarks

Call this method before calling [IAssemblyDoc::IGetFeatureScope](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::UpdateFeatureScope Method](#)
[IAssemblyDoc::RemoveFromFeatureScope Method](#)
[IAssemblyDoc::IGetFeatureScope Method](#)
[IAssemblyDoc::GetFeatureScope Method](#)
[IAssemblyDoc::AddToFeatureScope Method](#)
[IAssemblyDoc::IGetFeatureScope Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

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GetFirstMember Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetFirstMember Method (IAssemblyDoc)

Obsolete. Superseded by [IComponent2::GetChildren](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetFirstMember() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Object  
  
value = instance.GetFirstMember()
```

C#

```
System.object GetFirstMember()
```

C++/CLI

```
System.Object^ GetFirstMember();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetFirstMember](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

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GetLargeDesignReviewState Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetLargeDesignReviewState Method (IAssemblyDoc)

Gets the Large Design Review state of this assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLargeDesignReviewState() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Integer  
  
value = instance.GetLargeDesignReviewState()
```

C#

```
System.int GetLargeDesignReviewState()
```

C++/CLI

```
System.int GetLargeDesignReviewState();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Large Design Review state as defined by [swLargeDesignReviewState_e](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetLargeDesignReviewState](#).

Example

```
'VBA
'Open an assembly and run the macro.
'=====

Dim swApp As SldWorks.SldWorks
Dim swAssemDoc As SldWorks.AssemblyDoc
Dim boolstatus As Boolean
Dim longstatus As Long, longwarnings As Long
Dim LDRState As Long
Option Explicit
Sub main()
    Set swApp = Application.SldWorks
    Set swAssemDoc = swApp.ActiveDoc
    LDRState = swAssemDoc.GetLargeDesignReviewState
    If (LDRState = 0) Then
        Debug.Print "LDR State : None"
    ElseIf (LDRState = 1) Then
        Debug.Print "LDR State : LDR Mode"
    Else
        Debug.Print "LDR State : LDR with Assembly Edit Mode"
    End If
End Sub
```

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

Availability

SOLIDWORKS 2024 SP02, Revision Number 32.2

SOLIDWORKS API Help

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GetLightWeightComponentCount Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetLightWeightComponentCount Method (IAssemblyDoc)

Gets the number of lightweight components in the assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLightWeightComponentCount() As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IAssemblyDoc
Dim value As System.Integer
```

```
value = instance.GetLightWeightComponentCount()
```

C#



```
System.int GetLightWeightComponentCount()
```

C++/CLI

```
System.int GetLightWeightComponentCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of lightweight components

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetLightWeightComponentCount](#).

Example

[Make Assembly Components Lightweight \(VBA\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::LightweightAllResolved Method](#)
[IAssemblyDoc::ResolveAllLightweight Method](#)
[IAssemblyDoc::ResolveAllLightWeightComponents Method](#)
[IAssemblyDoc::ResolveOutOfDateLightWeightComponents Method](#)
[IAssemblyDoc::SetComponentSuppression Method](#)

SOLIDWORKS API Help

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GetPhysicalSimulationComponents Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetPhysicalSimulationComponents Method (IAssemblyDoc)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

```
Sub GetPhysicalSimulationComponents( _  
    ByVal InDuration As System.Double, _  
    ByRef OutCount As System.Integer, _  
    ByRef OutComponents As System.Object, _  
    ByRef OutTransforms As System.Object, _  
    ByRef OutStepStartTimes As System.Object, _  
    ByRef OutStepDurations As System.Object, _  
    ByRef OutTotalPhysSimDuration As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim InDuration As System.Double  
Dim OutCount As System.Integer  
Dim OutComponents As System.Object  
Dim OutTransforms As System.Object  
Dim OutStepStartTimes As System.Object  
Dim OutStepDurations As System.Object  
Dim OutTotalPhysSimDuration As System.Double
```

```
instance.GetPhysicalSimulationComponents(InDuration, OutCount, OutComponents, OutTransforms,  
OutStepStartTimes, OutStepDurations, OutTotalPhysSimDuration)
```

C#

```
void GetPhysicalSimulationComponents(  
    System.double InDuration,
```

```

    out System.int OutCount,
    out System.object OutComponents,
    out System.object OutTransforms,
    out System.object OutStepStartTimes,
    out System.object OutStepDurations,
    out System.double OutTotalPhysSimDuration
)

```

C++/CLI

```

void GetPhysicalSimulationComponents(
& System.double InDuration,
& [Out] System.int OutCount,
& [Out] System.Object^ OutComponents,
& [Out] System.Object^ OutTransforms,
& [Out] System.Object^ OutStepStartTimes,
& [Out] System.Object^ OutStepDurations,
& [Out] System.double OutTotalPhysSimDuration
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InDuration
Total elapsed time for Physical Simulation

OutCount
Size for all returned arrays

OutComponents
Array of the [IComponent2](#) objects

OutTransforms
Array of the [IMathTransform](#) objects

OutStepStartTimes
Array of doubles of size outCount; when each step in the Physical Simulation should happen

OutStepDurations
Array of doubles of size outCount; how long each step in the Physical Simulation should take

OutTotalPhysSimDuration
Total elapsed time Physical Simulation should have taken

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetPhysicalSimulationComponents](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[ISimulation Interface](#)
[ISimulationGravityFeatureData Interface](#)
[ISimulationLinearSpringFeatureData Interface](#)
[ISimulationMotorFeatureData Interface](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12

SOLIDWORKS API Help

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GetRouteManager Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetRouteManager Method (IAssemblyDoc)

Gets the SOLIDWORKS Routing API.

.NET Syntax

Visual Basic (Declaration)

Function GetRouteManager() As System.Object

Visual Basic (Usage)

```

Dim instance As IAssemblyDoc
Dim value As System.Object

```



```
value = instance.GetRouteManager()
```

C#

```
System.Object GetRouteManager()
```

C++/CLI

```
System.Object^ GetRouteManager();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

SOLIDWORKS Routing API

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetRouteManager](#).

Example

[Create Auto Route \(VBA\)](#)
[Create Auto Route \(VB.NET\)](#)
[Create Auto Route \(C#\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::IsRouteAssembly Method](#)
[IModelDoc2::EditRoute Method](#)
[IModelDoc2::InsertRoutePoint Method](#)
[IFeatureManager::InsertConnectionPoint Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

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GetSelectedMember Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetSelectedMember Method (IAssemblyDoc)

Obsolete. Superseded by [ISelectionMgr::GetSelectedObject6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSelectedMember() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Object  
  
value = instance.GetSelectedMember()
```

C#

```
System.Object GetSelectedMember()
```

C++/CLI

```
System.Object^ GetSelectedMember();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetSelectedMember](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSimulation Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetSimulation Method (IAssemblyDoc)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSimulation() As Simulation
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As Simulation  
  
value = instance.GetSimulation()
```

C#

```
Simulation GetSimulation()
```

C++/CLI

```
Simulation^ GetSimulation();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[ISimulation](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetSimulation](#).

Remarks

The **Simulation** folder is returned even if the folder does not yet exist or does not currently appear in the FeatureManager design tree.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

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GetUnloadedComponentNames Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetUnloadedComponentNames Method (IAssemblyDoc)

Gets the unloaded components' paths, referenced configuration names, reasons why they are unloaded, document types, and names.

.NET Syntax

Visual Basic (Declaration)

```
Function GetUnloadedComponentNames( _  
    ByRef UnloadedComponentPathNames As System.Object, _  
    ByRef UnloadedComponentReferencedConfigurationNames As System.Object, _  
    ByRef ReasonForUnloadingComponents As System.Object, _  
    ByRef DocTypes As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim UnloadedComponentPathNames As System.Object  
Dim UnloadedComponentReferencedConfigurationNames As System.Object  
Dim ReasonForUnloadingComponents As System.Object  
Dim DocTypes As System.Object  
Dim value As System.Object  
  
value = instance.GetUnloadedComponentNames(UnloadedComponentPathNames,  
UnloadedComponentReferencedConfigurationNames, ReasonForUnloadingComponents, DocTypes)
```

C#

```
System.object GetUnloadedComponentNames(  
    out System.object UnloadedComponentPathNames,  
    out System.object UnloadedComponentReferencedConfigurationNames,  
    out System.object ReasonForUnloadingComponents,  
    out System.object DocTypes  
)
```

C++/CLI

```
System.Object^ GetUnloadedComponentNames(  
& [Out] System.Object^ UnloadedComponentPathNames,  
& [Out] System.Object^ UnloadedComponentReferencedConfigurationNames,  
& [Out] System.Object^ ReasonForUnloadingComponents,  
& [Out] System.Object^ DocTypes  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UnloadedComponentPathNames

Array of the paths of the unloaded components

UnloadedComponentReferencedConfigurationNames

Array of the referenced configuration names of the unloaded components

ReasonForUnloadingComponents

Array indicating the reason each component is unloaded as defined by [swComponentLoadStatus_e](#)

DocTypes

Array of document types as defined in [swDocumentTypes_e](#)

Return Value

Array of the names of the unloaded components

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetUnloadedComponentNames](#).

Example

[Get Hidden Components Filenames \(C#\)](#)
[Get Hidden Components Filenames \(VB.NET\)](#)
[Get Hidden Components Filenames \(VBA\)](#)

Remarks

This method is useful when the assembly document was opened with [Quick View/Selective open](#) or [Do not load hidden components](#).
To get whether an assembly has hidden or suppressed unloaded components, call [IAssemblyDoc::HasUnloadedComponents](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IComponent2::HasUnloadedComponents Method](#)
[IComponent2::GetUnloadedComponentNames Method](#)
[IComponent2::GetHiddenUnloadedChildrenCount Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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GetVisibleComponentsInView Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetVisibleComponentsInView Method (IAssemblyDoc)

Gets a list of visible components in this assembly to save as solid bodies.

.NET Syntax

Visual Basic (Declaration)

```
Function GetVisibleComponentsInView() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Object
```

```
value = instance.GetVisibleComponentsInView()
```

C#

```
System.object GetVisibleComponentsInView()
```

C++/CLI

```
System.Object^ GetVisibleComponentsInView();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of visible [components](#) in this assembly

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetVisibleComponentsInView](#).

Remarks

One way to save an assembly as a multibody part document is to save the visible components in the assembly document as solid bodies. This method corresponds to the **Exterior Components** option in the **File, Save As** dialog box. For more information about saving assemblies as multibody documents, see SOLIDWORKS Help.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::IGetVisibleComponentsInView Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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GetVisibleComponentsInViewCount Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : GetVisibleComponentsInViewCount Method (IAssemblyDoc)

Gets the number of visible components in this assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function GetVisibleComponentsInViewCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Integer  
  
value = instance.GetVisibleComponentsInViewCount()
```

C#

```
System.int GetVisibleComponentsInViewCount()
```

C++/CLI

```
System.int GetVisibleComponentsInViewCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of visible components in this assembly

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::GetVisibleComponentsInViewCount](#).

Remarks

Call this method before calling [IAssembly_Doc::IGetVisibleComponentsInView](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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HasUnloadedComponents Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : HasUnloadedComponents Method (IAssemblyDoc)

Gets whether this assembly has hidden or suppressed unloaded components.

.NET Syntax

Visual Basic (Declaration)

```
Function HasUnloadedComponents() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim value As System.Boolean

value = instance.HasUnloadedComponents()
```

C#

```
System.bool HasUnloadedComponents()
```

C++/CLI

```
System.bool HasUnloadedComponents();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the assembly has hidden or suppressed unloaded components, false if not

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::HasUnloadedComponents](#).

Example

[Get Hidden Components Filenames \(C#\)](#)
[Get Hidden Components Filenames \(VB.NET\)](#)
[Get Hidden Component Filenames \(VBA\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::GetUnloadedComponentNames Method](#)
[IComponent2::GetHiddenUnloadedChildrenCount Method](#)
[IComponent2::GetUnloadedComponentNames Method](#)
[IComponent2::HasUnloadedComponents Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

HideComponent Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : HideComponent Method (IAssemblyDoc)

Obsolete. Superseded by [IModelDoc2::HideComponent2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub HideComponent()
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc

instance.HideComponent()
```

C#

```
void HideComponent()
```

C++/CLI

```
void HideComponent();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::HideComponent](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IAddComponent2 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IAddComponent2 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::AddComponent4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IAddComponent2( _  
    ByVal CompName As System.String, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As Component
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim CompName As System.String  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As Component  
  
value = instance.IAddComponent2(CompName, X, Y, Z)
```

C#

```
Component IAddComponent2(  
    System.string CompName,  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
Component^ IAddComponent2(  
& System.String^ CompName,  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CompName

X

Y

Z

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IAddComponent2](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IAddComponent3 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IAddComponent3 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::AddComponent4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IAddComponent3( _  
    ByVal CompName As System.String, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As Component2
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim CompName As System.String  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As Component2  
  
value = instance.IAddComponent3(CompName, X, Y, Z)
```

C#

```
Component2 IAddComponent3(  
    System.string CompName,  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
Component2^ IAddComponent3(  
& System.String^ CompName,  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CompName

X

Y

Z

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IAddComponent3](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IAddComponents Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IAddComponents Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::IAddComponents2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IAddComponents( _  
    ByRef Count As System.Integer, _  
    ByRef Names As System.String, _  
    ByRef Transforms As System.Double _  
    ) As Component
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Count As System.Integer  
Dim Names As System.String  
Dim Transforms As System.Double  
Dim value As Component
```

```
value = instance.IAddComponents(Count, Names, Transforms)
```

C#

```
Component IAddComponents(  
    ref System.int Count,  
    ref System.string Names,  
    ref System.double Transforms  
)
```

C++/CLI

```
Component^ IAddComponents(  
& System.int% Count,  
& System.String^% Names,  
& System.double% Transforms  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Names

Transforms

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IAddComponents](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IAddComponents2 Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IAddComponents2 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::IAddComponents3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IAddComponents2( _  
    ByRef Count As System.Integer, _  
    ByRef Names As System.String, _  
    ByRef Transforms As System.Double _  
) As Component2
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Count As System.Integer  
Dim Names As System.String  
Dim Transforms As System.Double  
Dim value As Component2  
  
value = instance.IAddComponents2(Count, Names, Transforms)
```

C#

```
Component2 IAddComponents2(  
    ref System.int Count,  
    ref System.string Names,  
    ref System.double Transforms  
)
```

C++/CLI

```
Component2^ IAddComponents2(  
& System.int% Count,  
& System.String^% Names,  
& System.double% Transforms  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of components to add

Names

- in-process, unmanaged C++: Pointer to an array of strings of component file names
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Transforms

- in-process, unmanaged C++: Pointer to an array of doubles of component transforms
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Return Value

- in-process, unmanaged C++: Pointer to an array of [IComponent2](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IAddComponents2](#).

Example

[Add Components and Transforms \(C++\)](#)

Remarks

The array of file names represents all of the components that you want to add to the assembly. If there is more than one instance of a given component, make sure to specify the file name for each instance of the component that you want to add.

The array of transforms consists of [*count* x 16] doubles. There should be one transform for each component to be added.

The application must allocate the array of components, and it is also responsible for releasing the returned array of components.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::AddComponents Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IAddComponents3 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IAddComponents3 Method (IAssemblyDoc)

Adds multiple components to the assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function IAddComponents3( _  
    ByVal Count As System.Integer, _  
    ByRef Names As System.String, _  
    ByRef Transforms As System.Double, _  
    ByVal CoordinateSystemNameCount As System.Integer, _  
    ByRef CoordinateSystemNames As System.String _  
    ) As Component2
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Count As System.Integer  
Dim Names As System.String  
Dim Transforms As System.Double  
Dim CoordinateSystemNameCount As System.Integer  
Dim CoordinateSystemNames As System.String  
Dim value As Component2  
  
value = instance.IAddComponents3(Count, Names, Transforms, CoordinateSystemNameCount,  
CoordinateSystemNames)
```

C#

```
Component2 IAddComponents3(  
    System.int Count,  
    ref System.string Names,  
    ref System.double Transforms,  
    System.int CoordinateSystemNameCount,  
    ref System.string CoordinateSystemNames  
)
```

C++/CLI

```
Component2^ IAddComponents3(  
& System.int Count,  
& System.String^% Names,  
& System.double% Transforms,  
& System.int CoordinateSystemNameCount,  
& System.String^% CoordinateSystemNames  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of components to add

Names

- in-process, unmanaged C++: Pointer to an array of strings of full path names of components to add
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Transforms

- in-process, unmanaged C++: Pointer to an array of doubles of component transforms
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

CoordinateSystemNameCount

Number of coordinate system names in CoordinateSystemNames; include in the count all empty strings

CoordinateSystemNames

- in-process, unmanaged C++: Pointer to an array of strings of coordinate system names
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

- in-process, unmanaged C++: Pointer to an array of [IComponent2](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IAddComponents3](#).

Remarks

Transforms contains an array of [Count x 16] doubles. This parameter stores one transformation matrix of 16 doubles for each component in Names. If a component's transformation matrix is null, then the component is placed in the assembly such that the component's user-defined coordinate system coincides exactly with the default coordinate system of the assembly (no transformation). See [IMathTransform](#) for details about transformation matrices.

CoordinateSystemNames contains a user-defined coordinate system name for each component in Names. If a component's user-defined coordinate system is an empty string, then the component is placed in the assembly with respect to the default coordinate system of the component.

The application must allocate the array of components, and it is also responsible for releasing the returned array of components.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::AddComponents3 Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IComponentReload2 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IComponentReload2 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::ReplaceComponents](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IComponentReload2( _  
    ByVal Component As Component, _  
    ByVal ReadOnly As System.Boolean, _  
    ByVal Options As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim Component As Component
Dim ReadOnly As System.Boolean
Dim Options As System.Integer
Dim value As System.Integer

value = instance.IComponentReload2(Component, ReadOnly, Options)
```

C#

```
System.int IComponentReload2(
    Component Component,
    System.bool ReadOnly,
    System.int Options
)
```

C++/CLI

```
System.int IComponentReload2(
    & Component^ Component,
    & System.bool ReadOnly,
    & System.int Options
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Component

ReadOnly

Options

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc.IComponentReload2](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IComponentReload3 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IComponentReload3 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc.ReplaceComponents](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IComponentReload3( _
    ByVal Component As Component2, _
    ByVal ReadOnly As System.Boolean, _
    ByVal Options As System.Integer _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim Component As Component2
Dim ReadOnly As System.Boolean
Dim Options As System.Integer
Dim value As System.Integer
```

```
value = instance.IComponentReload3(Component, ReadOnly, Options)
```

C#

```
System.int IComponentReload3(  
    Component2 Component,  
    System.bool ReadOnly,  
    System.int Options  
)
```

C++/CLI

```
System.int IComponentReload3(  
& Component2^ Component,  
& System.bool ReadOnly,  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Component

ReadOnly

Options

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IComponentReload3](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IFeatureByName Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IFeatureByName Method (IAssemblyDoc)

Returns the [IFeature](#) object for the named feature in the assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function IFeatureByName(  
    ByVal Name As System.String _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Name As System.String  
Dim value As Feature
```

```
value = instance.IFeatureByName(Name)
```

C#

```
Feature IFeatureByName(  
    System.string Name  
)
```

C++/CLI

```
Feature^ IFeatureByName(  
& System.String^ Name  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name
Name of the feature

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IFeatureByName](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::FeatureByName Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetBox Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IGetBox Method (IAssemblyDoc)

Gets the bounding box.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetBox( _  
    ByVal Options As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Options As System.Integer  
Dim value As System.Double
```

```
value = instance.IGetBox(Options)
```

C#

```
System.double IGetBox(  
    System.int Options  
)
```

C++/CLI

```
System.double IGetBox(  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Options
Type of bounding box as defined by [swBoundingBoxOptions_e](#)

Return Value

Two diagonal points that bound the object; the format is a pointer to an array of 6 doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IGetBox](#).

Remarks

IMPORTANT: The values returned are approximate and should not be used for comparison or calculation purposes. Furthermore, the bounding box may vary after rebuilding the model. The resulting box encloses the object, but it might not be the tightest box.

The X, Y, Z points returned by SOLIDWORKS are the lower- and upper-diagonal corners that bound the component with the box sides parallel to the X, Y and Z axes. SOLIDWORKS returns box dimensions that enclose the object and are typically close to the minimum possible size.

The return value is an array of doubles as follows:

[XCorner1, YCorner1, ZCorner1, XCorner2, YCorner2, ZCorner2]

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::GetBox Method](#)
[IComponent2::GetBox Method](#)
[IModelDocExtension::GetVisibleBox Method](#)
[IModelDocExtension::RemoveVisibleBox Method](#)
[IModelDocExtension::SetVisibleBox Method](#)
[IAssemblyDoc::UpdateBox Method](#)
[IFeature::GetBox Method](#)
[IFeature::IGetBox Method](#)
[IFace2::GetBox Method](#)
[IFace2::IGetBox Method](#)
[IComponent2::IGetBox Method](#)
[IBody2::GetBodyBox Method](#)
[IBody2::IGetBodyBox Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetComponents Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IGetComponents Method (IAssemblyDoc)

Gets all of the components in the active configuration of this assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetComponents( _  
    ByVal ToplevelOnly As System.Boolean, _  
    ByVal Count As System.Integer _  
) As Component2
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim ToplevelOnly As System.Boolean  
Dim Count As System.Integer  
Dim value As Component2
```

```
value = instance.IGetComponents(ToplevelOnly, Count)
```

C#

```
Component2 IGetComponents(  
    System.bool ToplevelOnly,  
    System.int Count  
)
```

C++/CLI

```
Component2^ IGetComponents(  
& System.bool TopLevelOnly,  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopLevelOnly

True to get only the number of components at the top level of the FeatureManager design tree, false to get the number of top level and all child components in the FeatureManager design tree

Count

Number of components

Return Value

- in-process, unmanaged C++: Pointer to an array of [IComponent2](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before using this method, use [IAssemblyDoc::GetComponentCount](#) to determine the size of the array to pass to this method.

The components returned by this method can be in any order. You should not rely on the order to indicate anything about children or parents. If the hierarchy and order are important, then use [IModelDoc2::FirstFeature](#), [IFeature::GetTypeName2](#), [IFeature::GetSpecificFeature2](#), [IComponent2::FirstFeature](#), and [IFeature::GetNextFeature](#) to retrieve your information.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IComponent2::GetChildren Method](#)
[IComponent2::GetParent Method](#)
[IConfiguration::GetRootComponent3 Method](#)
[IConfiguration::IGetRootComponent2 Method](#)
[IConfigurationManager::ActiveConfiguration Property](#)
[IAssemblyDoc::GetComponents Method](#)
[IAssemblyDoc::GetComponentByID Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDragOperator Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IGetDragOperator Method (IAssemblyDoc)

Gets the drag operator for dynamic drag operations in this assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDragOperator() As DragOperator
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As DragOperator
```

```
value = instance.IGetDragOperator()
```

C#

```
DragOperator IGetDragOperator()
```

C++/CLI

```
DragOperator^ IGetDragOperator();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IDragOperator](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IGetDragOperator](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::GetDragOperator Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetEditTarget Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IGetEditTarget Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::IGetEditTarget2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEditTarget() As ModelDoc
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As ModelDoc
```

```
value = instance.IGetEditTarget()
```

C#

```
ModelDoc IGetEditTarget()
```

C++/CLI

```
ModelDoc^ IGetEditTarget();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IGetEditTarget](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetEditTarget2 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IGetEditTarget2 Method (IAssemblyDoc)

Gets the model document that is currently being edited.

.NET Syntax

Visual Basic (Declaration)

Function IGetEditTarget2() As ModelDoc2

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim value As ModelDoc2

value = instance.IGetEditTarget2()
```

C#

```
ModelDoc2 IGetEditTarget2()
```

C++/CLI

```
ModelDoc2^ IGetEditTarget2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IModelDoc2](#) being edited

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IGetEditTarget2](#).

Remarks

This method is useful when the user is performing an in-context edit of a part component because the currently active document returned from [ISldWorks::ActiveDoc](#) might not be the document that the user is editing. For example, the user can use in-context editing to modify an assembly component. The active document is the assembly, but this method returns the IModelDoc2 for the component being edited.

The IModelDoc2 object returned by this method might be that of the assembly if no part is being in-place edited. In other words, this method returns a valid pointer (non-NULL) even when user is not editing a part in-context.

You can use the IModelDoc2 object returned by this method to determine which assembly component is being edited in-context. In general, you should not use this IModelDoc2 object for feature creation within the component part. Feature creation typically requires the IModelDoc2 of the active document. During feature creation, SOLIDWORKS automatically determines whether the feature should be created and owned by the active assembly, or if it is an in-context edit in which the feature should be created and owned by the component part.

COM applications need to release the IModelDoc2 object returned because this method increments the reference count on the IModelDoc2 object even if the edit target is the active document.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IModelDoc2::IsEditingSelf Method](#)
[IAssemblyDoc::GetEditTarget Method](#)
[IAssemblyDoc::GetEditTargetComponent Method](#)
[ISelectionMgr::IsInEditTarget2 Method](#)
[ISldWorks::IActiveDoc2 Property](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetExplodedViewNames Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IGetExplodedViewNames Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::GetExplodedViewNames2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetExplodedViewNames( _  
    ByVal Count As System.Integer _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim Count As System.Integer
Dim value As System.String

value = instance.IGetExplodedViewNames(Count)
```

C#

```
System.string IGetExplodedViewNames(
    System.int Count
)
```

C++/CLI

```
System.String^ IGetExplodedViewNames(
    & System.int Count
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of exploded views in the active configuration (see **Remarks**)

Return Value

- in-process, unmanaged C++: Pointer to an array of strings of the names of the exploded views in the active configuration
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IAssemblyDoc::GetExplodedViewCount](#) to get the value of Count.

To take advantage of the explode mechanism, see [IAssemblyDoc::CreateExplodedView](#), [IAssemblyDoc::GetExplodedViewNames](#), [IAssemblyDoc::GetExplodedViewCount](#), [IAssemblyDoc::AutoExplode](#), [IAssemblyDoc::ShowExploded2](#), [IAssemblyDoc::ViewExplodeAssembly](#), [IAssemblyDoc::ViewCollapseAssembly](#), [IView::ShowExploded](#), [IView::IsExploded](#), and [IModelDoc2::IsExploded](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFeatureScope Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IGetFeatureScope Method (IAssemblyDoc)

Gets the components affected by this feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFeatureScope( _
    ByVal FeatureIn As Feature, _
    ByVal NumComponents As System.Integer _
) As Component2
```

Visual Basic (Usage)

```
Dim instance As IAssemblyDoc
Dim FeatureIn As Feature
Dim NumComponents As System.Integer
```



```
Dim value As Component2
```

```
value = instance.IGetFeatureScope(FeatureIn, NumComponents)
```

C#

```
Component2 IGetFeatureScope(  
    Feature FeatureIn,  
    System.int NumComponents  
)
```

C++/CLI

```
Component2^ IGetFeatureScope(  
& Feature^ FeatureIn,  
& System.int NumComponents  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatureIn

[IFeature](#)

NumComponents

Number of components affected by this feature

Return Value

Array of [components](#) of size NumComponents

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IGetFeatureScope](#).

Remarks

Feature scope information is only provided if the assembly is opened in its own window.

COM users must call [IAssemblyDoc::GetFeatureScopeCount](#) before calling this method.

This method gets both the components affected and potentially affected by this feature. Thus, not all of the components in this list are necessarily affected by this feature.

See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[IAssemblyDoc::GetFeatureScope Method](#)

[IAssemblyDoc::AddToFeatureScope Method](#)

[IAssemblyDoc::GetFeatureScope Method](#)

[IAssemblyDoc::GetFeatureScopeCount Method](#)

[IAssemblyDoc::UpdateFeatureScope Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFirstMember Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IGetFirstMember Method (IAssemblyDoc)

Obsolete. Superseded by [IComponent2::GetChildren](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstMember() As Member
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As Member  
  
value = instance.IGetFirstMember()
```

C#

```
Member IGetFirstMember()
```

C++/CLI

```
Member^ IGetFirstMember();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IGetFirstMember](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSelectedMember Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IGetSelectedMember Method (IAssemblyDoc)

Obsolete. Superseded by [SelectionMgr::GetSelectedObject6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSelectedMember() As Member
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As Member  
  
value = instance.IGetSelectedMember()
```

C#

```
Member IGetSelectedMember()
```

C++/CLI

```
Member^ IGetSelectedMember();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IGetSelectedMember](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetVisibleComponentsInView Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IGetVisibleComponentsInView Method (IAssemblyDoc)

Gets a list of visible components in this assembly to save as solid bodies.

VB.NET Syntax

Visual Basic (Declaration)

```
Function IGetVisibleComponentsInView( _  
    ByVal Count As System.Integer _  
) As Component2
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Count As System.Integer  
Dim value As Component2  
  
value = instance.IGetVisibleComponentsInView(Count)
```

C#

```
Component2 IGetVisibleComponentsInView(  
    System.int Count  
)
```

C++/CLI

```
Component2^ IGetVisibleComponentsInView(  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of visible components in this assembly

Return Value

- in-process, unmanaged C++: Pointer to an array of visible [components](#) in this assembly of size count
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

One way to save an assembly as a multibody part document is to save the visible components in the assembly document as solid bodies. This method corresponds to the **Exterior Components** option in the **File, Save As** dialog box. For more information about saving assemblies as multibody documents, see SOLIDWORKS Help.

Call [IAssemblyDoc::GetVisibleComponentsInViewCount](#) before calling this method.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::GetVisibleComponentsInView Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMirrorComponents Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : IMirrorComponents Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::MirrorComponents2](#).

Visual Basic (Declaration)

```
Function IMirrorComponents( _
    ByVal Plane As System.Object, _
    ByVal InstanceCount As System.Integer, _
    ByRef ComponentsToInstance As Component2, _
    ByVal MirrorCount As System.Integer, _
    ByRef ComponentsToMirror As Component2, _
    ByVal NameCount As System.Integer, _
    ByRef MirroredComponentFileNames As System.String, _
    ByVal RecreateMates As System.Boolean, _
    ByVal ComponentModifier As System.Integer, _
    ByVal ComponentNameModifier As System.String, _
    ByVal MirroredFileLocation As System.String, _
    ByVal CopyCustomProperties As System.Boolean _
) As Component2
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim Plane As System.Object
Dim InstanceCount As System.Integer
Dim ComponentsToInstance As Component2
Dim MirrorCount As System.Integer
Dim ComponentsToMirror As Component2
Dim NameCount As System.Integer
Dim MirroredComponentFileNames As System.String
Dim RecreateMates As System.Boolean
Dim ComponentModifier As System.Integer
Dim ComponentNameModifier As System.String
Dim MirroredFileLocation As System.String
Dim CopyCustomProperties As System.Boolean
Dim value As Component2
```

```
value = instance.IMirrorComponents(Plane, InstanceCount, ComponentsToInstance, MirrorCount,
ComponentsToMirror, NameCount, MirroredComponentFileNames, RecreateMates, ComponentModifier,
ComponentNameModifier, MirroredFileLocation, CopyCustomProperties)
```

C#

```
Component2 IMirrorComponents(
    System.object Plane,
    System.int InstanceCount,
    ref Component2 ComponentsToInstance,
    System.int MirrorCount,
    ref Component2 ComponentsToMirror,
    System.int NameCount,
    ref System.string MirroredComponentFileNames,
    System.bool RecreateMates,
    System.int ComponentModifier,
    System.string ComponentNameModifier,
    System.string MirroredFileLocation,
    System.bool CopyCustomProperties
)
```

C++/CLI

```
Component2^ IMirrorComponents(
    & System.Object^ Plane,
    & System.int InstanceCount,
    & Component2^% ComponentsToInstance,
    & System.int MirrorCount,
    & Component2^% ComponentsToMirror,
    & System.int NameCount,
    & System.String^% MirroredComponentFileNames,
    & System.bool RecreateMates,
    & System.int ComponentModifier,
```

```
& System.String^ ComponentNameModifier,  
& System.String^ MirroredFileLocation,  
& System.bool CopyCustomProperties  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Plane
Plane or planar face about which to mirror the components
InstanceCount
Number of instances of the components to mirror
ComponentsToInstance
Array of instances of the [component](#) to mirror
MirrorCount
Number of components to mirror
ComponentsToMirror
Array of [components](#) to mirror
NameCount
Number of filenames for the newly created mirrored assemblies or parts
MirroredComponentFilenames
Array of filenames for the newly created mirrored assemblies or parts
RecreateMates
True to preserve any mates between the selected components if more than one component is to be mirrored, false to not
ComponentModifier
Prefix or suffix for the newly mirrored components if *MirroredComponentFilenames* is not specified, as defined by [swMirrorComponentNameModifier_e](#)
ComponentNameModifier
String to add to the prefix or suffix of the name of the newly mirrored components if *MirroredComponentFilenames* is not specified
MirroredFileLocation
Name of the folder where to store the file of the newly created mirrored components
CopyCustomProperties
True to copy the custom properties from the selected components to the mirrored components, false to not

Return Value

- in-process, unmanaged C++: Pointer to an array of newly created [components](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::IMirrorComponents](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::MirrorComponents Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertCavity Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertCavity Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::InsertCavity4](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertCavity()
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
```

```
instance.InsertCavity()
```

C#

```
void InsertCavity()
```

C++/CLI

```
void InsertCavity();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::InsertCavity](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertCavity2 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertCavity2 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::InsertCavity4](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertCavity2( _  
    ByVal ScaleFactor As System.Double, _  
    ByVal ScaleType As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim ScaleFactor As System.Double  
Dim ScaleType As System.Integer  
  
instance.InsertCavity2(ScaleFactor, ScaleType)
```

C#

```
void InsertCavity2(  
    System.double ScaleFactor,  
    System.int ScaleType  
)
```

C++/CLI

```
void InsertCavity2(  
& System.double ScaleFactor,  
& System.int ScaleType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ScaleFactor

ScaleType

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::InsertCavity2](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

InsertCavity3 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertCavity3 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::InsertCavity4](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertCavity3( _  
    ByVal ScaleFactor As System.Double, _  
    ByVal ScaleType As System.Integer, _  
    ByVal KeepPieceIndex As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim ScaleFactor As System.Double  
Dim ScaleType As System.Integer  
Dim KeepPieceIndex As System.Integer  
  
instance.InsertCavity3(ScaleFactor, ScaleType, KeepPieceIndex)
```

C#

```
void InsertCavity3(  
    System.double ScaleFactor,  
    System.int ScaleType,  
    System.int KeepPieceIndex  
)
```

C++/CLI

```
void InsertCavity3(  
& System.double ScaleFactor,  
& System.int ScaleType,  
& System.int KeepPieceIndex  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ScaleFactor

ScaleType

KeepPieceIndex

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::InsertCavity3](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

InsertCavity4 Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertCavity4 Method (IAssemblyDoc)

Inserts a cavity to the active part using a selected component.

Visual Basic (Declaration)

```
Sub InsertCavity4( _
    ByVal ScaleFactor\_x As System.Double, _
    ByVal ScaleFactor\_y As System.Double, _
    ByVal ScaleFactor\_z As System.Double, _
    ByVal IsUniform As System.Boolean, _
    ByVal ScaleType As System.Integer, _
    ByVal KeepPieceIndex As System.Integer _
)
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim ScaleFactor_x As System.Double
Dim ScaleFactor_y As System.Double
Dim ScaleFactor_z As System.Double
Dim IsUniform As System.Boolean
Dim ScaleType As System.Integer
Dim KeepPieceIndex As System.Integer

instance.InsertCavity4(ScaleFactor_x, ScaleFactor_y, ScaleFactor_z, IsUniform, ScaleType,
KeepPieceIndex)
```

C#

```
void InsertCavity4(
    System.double ScaleFactor\_x,
    System.double ScaleFactor\_y,
    System.double ScaleFactor\_z,
    System.bool IsUniform,
    System.int ScaleType,
    System.int KeepPieceIndex
)
```

C++/CLI

```
void InsertCavity4(
    & System.double ScaleFactor\_x,
    & System.double ScaleFactor\_y,
    & System.double ScaleFactor\_z,
    & System.bool IsUniform,
    & System.int ScaleType,
    & System.int KeepPieceIndex
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ScaleFactor_x

Scaling factor in the x direction

ScaleFactor_y

Scaling factor in the y direction

ScaleFactor_z

Scaling factor in the z direction

IsUniform

True to use the first scale argument as the uniform scale, false to not

ScaleType

Type of scaling as defined in [swCavityScaleType_e](#)

KeepPieceIndex

Piece to keep if there is ambiguity

Example

[Insert Cavity \(C#\)](#)
[Insert Cavity \(VB.NET\)](#)
[Insert Cavity \(VBA\)](#)

Remarks

This operation is performed in the context of an assembly document. The component being edited in the context of the assembly receives the new cavity feature.

Set the scaleFactor argument as appropriate for your casting material. The scaling factor is expressed as a percentage (+/- 20%) of the size of the cavity part. Pass it in as a value within the range of -20 to +20.

SOLIDWORKS uses the following formula to determine the size of the cavity:

$$\text{cavitysize} = \text{partsize} * (1 + \text{scaleFactor}/100)$$

When there is ambiguity in the result of a cut, SOLIDWORKS uses KeepPieceIndex to determine which result to use. You can set this parameter to -1 if there is no ambiguity; otherwise, you should use the index of the result, in the range between 0 and 1 less than the possible number of outcomes.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[ICavityFeatureData Interface](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertDerivedPattern Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertDerivedPattern Method (IAssemblyDoc)

Obsolete. See [IFeatureManager::CreateFeature](#) and the Remarks in [IDerivedPatternFeatureData](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertDerivedPattern() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Boolean
```

```
value = instance.InsertDerivedPattern()
```

C#

```
System.bool InsertDerivedPattern()
```

C++/CLI

```
System.bool InsertDerivedPattern();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the derived component from the selected pattern and seed components is created, false if not

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc.InsertDerivedPattern](#).

Example

[Get Number of Instances Skipped in Driving Feature \(C#\)](#)
[Get Number of Instances Skipped in Driving Feature \(VB.NET\)](#)
[Get Number of Instances Skipped in Driving Feature \(VBA\)](#)

Remarks

Use [IModelDocExtension::SelectByID2](#) to select the components, which must be ordered:

- seed component = 1
- pattern feature = 6

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

Availability

SOLIDWORKS 2003 SP4, Revision Number 11.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertEnvelope Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertEnvelope Method (IAssemblyDoc)

Adds an envelope in the specified configuration name in this assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertEnvelope( _  
    ByVal CompName As System.String, _  
    ByVal ConfigName As System.String, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As Component2
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim CompName As System.String  
Dim ConfigName As System.String  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As Component2
```

```
value = instance.InsertEnvelope(CompName, ConfigName, X, Y, Z)
```

C#

```
Component2 InsertEnvelope(  
    System.string CompName,  
    System.string ConfigName,  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
Component2^ InsertEnvelope(  
& System.String^ CompName,  
& System.String^ ConfigName,
```

```
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CompName

Path and file name of file containing a part to insert as an envelope

ConfigName

Name of the configuration from which to load the envelope component

X

X coordinate of the center of the envelope component's bounding box

Y

Y coordinate of the center of the envelope component's bounding box

Z

Z coordinate of the center of the envelope component's bounding box

Return Value

[IComponent2](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::InsertEnvelope](#).

Remarks

CompName does not have to be open before running this method. If it is not open, then this method opens it in a hidden state.

If ConfigName is empty or specifies a configuration that does not exist, then the current configuration is used.

The X, Y, Z parameters of this method are the center of the component bounding box. You must use [IComponent2::Transform2](#) on the return interface pointer if you want to more precisely position the component.

See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[IComponent2::ISEnvelope Method](#)

[IAssemblyDoc::AddComponent5 Method](#)

[IAssemblyDoc::AddComponents Method](#)

[IAssemblyDoc::IAddComponents2 Method](#)

[IAssemblyDoc::ReplaceComponents Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertImportedComponent Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertImportedComponent Method (IAssemblyDoc)

Inserts a third-party native CAD part or assembly into the current configuration of this assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertImportedComponent( _  
    ByVal FileName As System.String, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double, _  
    ByRef CompInserted As System.Object _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IAssemblyDoc  
Dim FileName As System.String
```



```
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim CompInserted As System.Object
Dim value As System.Integer
```

```
value = instance.InsertImportedComponent(FileName, X, Y, Z, CompInserted)
```

C#

```
System.int InsertImportedComponent(
    System.string FileName,
    System.double X,
    System.double Y,
    System.double Z,
    out System.object CompInserted
)
```

C++/CLI

```
System.int InsertImportedComponent(
    & System.String^ FileName,
    & System.double X,
    & System.double Y,
    & System.double Z,
    & [Out] System.Object^ CompInserted
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Full path name of the third-party native CAD file to insert

X

X-coordinate of the component center

Y

Y-coordinate of the component center

Z

Z-coordinate of the component center

CompInserted

[IComponent2](#); null or Nothing if unsuccessful

Return Value

Error code as defined in [sw3DInterconnectImportErrors_e](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::InsertImportedComponent](#).

Remarks

To successfully use this method, you must first turn on 3D Interconnect by either:

- Setting **Tools > Options > System Options > Import > Enable 3D Interconnect**
- OR -
- Calling [ISldWorks::SetUserPreferenceToggle](#)([swUserPreferenceToggle_e](#).swMultiCAD_Enable3DInterconnect, True)

To edit the imported assembly, save the SOLIDWORKS assembly and then use [IFeature::GetImportedFeatureParameters](#) and [IFeature::SetImportedFeatureParameters](#).

See the [IImport3DInterconnectData](#) Remarks.

If 3D Interconnect is turned off, use [IAssemblyDoc::AddComponent5](#) instead of this method.

See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[ISldWorks::LoadFile4 Method \(\)](#)

[IPartDoc::InsertImportedFeature Method \(\)](#)

Availability

InsertJoin Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertJoin Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::InsertJoin2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertJoin()
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
```

```
instance.InsertJoin()
```

C#

```
void InsertJoin()
```

C++/CLI

```
void InsertJoin();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::InsertJoin](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

InsertJoin2 Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertJoin2 Method (IAssemblyDoc)

Constructs a feature from merged selected components.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertJoin2(  
    ByVal HideParts As System.Boolean, _  
    ByVal ForceContact As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
```

```
Dim HideParts As System.Boolean
```

```
Dim ForceContact As System.Boolean
```

```
Dim value As System.Boolean
```

```
value = instance.InsertJoin2(HideParts, ForceContact)
```

C#

```
System.bool InsertJoin2(  
    System.bool HideParts,
```

```
System.bool ForceContact
)
```

C++/CLI

```
System.bool InsertJoin2(
& System.bool HideParts,
& System.bool ForceContact
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

HideParts

True hides the original components after the join is complete, false shows them

ForceContact

True joins any coincident faces and intruding volumes, false does not

Return Value

True if feature creation was successful, false if it was not

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::InsertJoin2](#).

Example

[Insert Join Feature \(C#\)](#)
[Insert Join Feature \(VB.NET\)](#)
[Insert Join Feature \(VBA\)](#)

Remarks

When you use this method, SOLIDWORKS must be in Edit Part mode for the target part. You must also preselect the component parts to be merged into the edited part.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IJoinFeatureData Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertLoadReference Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertLoadReference Method (IAssemblyDoc)

Creates a mate load reference to the specified or selected mate.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertLoadReference( _
    ByVal Mate As Mate2 _
) As MateLoadReference
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim Mate As Mate2
Dim value As MateLoadReference

value = instance.InsertLoadReference(Mate)
```

C#

```
MateLoadReference InsertLoadReference(
    Mate2 Mate
)
```

C++/CLI

```
MateLoadReference^ InsertLoadReference(
    & Mate2^ Mate
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Mate

[Mate](#) to which to add a mate load reference; if NULL passed in, then the mate must already be selected

Return Value

[IMateLoadReference](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::InsertLoadReference](#).

Example

[Insert Mate Load Reference \(C#\)](#)
[Insert Mate Load Reference \(VB.NET\)](#)
[Insert Mate Load Reference \(VBA\)](#)

Remarks

The supplemental faces for the mate load reference must be selected before using this method. SOLIDWORKS determines which components own the selected supplemental faces and matches them to the components associated with the specified or selected mate. If the component of selected supplemental face does not match either of the components of the mate, then that face is ignored.

This method rebuilds the FeatureManager design tree, which can be a time-consuming operation if the FeatureManager design tree is large. If using this method to add many load references at once, use [IFeatureManager::EnableFeatureTree](#) before and after using [IAssemblyDoc::InsertLoadReference](#) to disable and then re-enable rebuilding the FeatureManager design tree.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertNewAssembly Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertNewAssembly Method (IAssemblyDoc)

Creates a new virtual sub-assembly and optionally saves it to the specified file.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertNewAssembly( _  
    ByVal FileName As System.String _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim FileName As System.String  
Dim value As System.Integer  
  
value = instance.InsertNewAssembly(FileName)
```

C#

```
System.int InsertNewAssembly(  
    System.string FileName  
)
```

C++/CLI

```
System.int InsertNewAssembly(  
& System.String^ FileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Full pathname of new sub-assembly document (see [Remarks](#))

Return Value

Error code as defined by [swInsertNewAssemblyErrorCode_e](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::InsertNewAssembly](#).

Example

[Insert and Save Virtual Assembly \(C#\)](#)
[Insert and Save Virtual Assembly \(VB.NET\)](#)
[Insert and Save Virtual Assembly \(VBA\)](#)

Remarks

If **Tools > Options > System Options > Assemblies > Save new components to external files** is:

- selected, then this API creates a virtual sub-assembly and saves it to the file specified in *FileName*.
- not selected, then this API ignores *FileName*, does not save the sub-assembly, and creates only a virtual sub-assembly in the FeatureManager design tree.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::InsertNewVirtualAssembly Method](#)
[IComponent2::SaveVirtualComponent Method](#)
[IComponent2::IsVirtual Property](#)
[IModelDocExtension::IsVirtualComponent3 Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertNewPart Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertNewPart Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::InsertNewPart2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertNewPart()
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
instance.InsertNewPart()
```

C#

```
void InsertNewPart()
```

C++/CLI

```
void InsertNewPart();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::InsertNewPart](#).

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertNewPart2 Method (IAssemblyDoc)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertNewPart2 Method (IAssemblyDoc)

Inserts a new part on the specified face or plane.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertNewPart2( _  
    ByVal FilePathIn As System.String, _  
    ByVal Face\_or\_Plane\_to\_select As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim FilePathIn As System.String  
Dim Face_or_Plane_to_select As System.Object  
Dim value As System.Integer  
  
value = instance.InsertNewPart2(FilePathIn, Face_or_Plane_to_select)
```

C#

```
System.int InsertNewPart2(  
    System.string FilePathIn,  
    System.object Face\_or\_Plane\_to\_select  
)
```

C++/CLI

```
System.int InsertNewPart2(  
& System.String^ FilePathIn,
```

```
& System.Object^ Face or Plane to select  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FilePathIn

Path and filename for the new part

Face_or_Plane_to_select

[Face](#) or [reference plane](#) where to insert the new part

Return Value

Error code as defined by [swInsertNewPartErrorCode_e](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::InsertNewPart2](#).

Example

[Insert Join Feature \(C#\)](#)

[Insert Join Feature \(VB.NET\)](#)

[Insert Join Feature \(VBA\)](#)

Remarks

The Face_or_Plane_to_select argument can be a [face](#) or [reference plane](#), which means that you have to call [IFeature::GetSpecificFeature2](#) first.

See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[IPartDoc Interface](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertNewVirtualAssembly Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertNewVirtualAssembly Method (IAssemblyDoc)

Creates a new assembly from this assembly and saves it internally as a virtual component.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertNewVirtualAssembly( _  
    ByRef InsertedComponent As Component2 _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim InsertedComponent As Component2  
Dim value As System.Integer  
  
value = instance.InsertNewVirtualAssembly(InsertedComponent)
```

C#

```
System.int InsertNewVirtualAssembly(  
    out Component2 InsertedComponent  
)
```

C++/CLI

```
System.int InsertNewVirtualAssembly(  
& [Out] Component2^ InsertedComponent  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InsertedComponent
New assembly inserted as virtual component

Return Value

Error code as defined by [swInsertNewPartErrorCode_e](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::InsertNewVirtualAssembly](#).

Example

[Insert New Virtual Assembly \(VBA\)](#)
[Insert New Virtual Assembly \(VB.NET\)](#)
[Insert New Virtual Assembly \(C#\)](#)

Remarks

If nothing is pre-selected, this method inserts the virtual assembly into the main assembly. If a sub-assembly is pre-selected, it inserts the virtual assembly into the sub-assembly.

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IAssemblyDoc::InsertNewVirtualPart Method](#)
[IComponent2::IsVirtual Property](#)
[IModelDocExtension::IsVirtualComponent3 Method](#)
[IAssemblyDoc::InsertNewAssembly Method](#)
[IComponent2::MakeVirtual Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertNewVirtualPart Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : InsertNewVirtualPart Method (IAssemblyDoc)

Creates a new part in the context of an assembly and saves the part internally in the assembly file as a virtual component.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertNewVirtualPart( _  
    ByVal FaceOrPlaneToSelect As System.Object, _  
    ByRef InsertedComponent As Component2 _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IAssemblyDoc  
Dim FaceOrPlaneToSelect As System.Object  
Dim InsertedComponent As Component2  
Dim value As System.Integer
```

```
value = instance.InsertNewVirtualPart(FaceOrPlaneToSelect, InsertedComponent)
```

C#



```
System.int InsertNewVirtualPart(  
    System.object FaceOrPlaneToSelect,  
    out Component2 InsertedComponent  
)
```

C++/CLI

```
System.int InsertNewVirtualPart(  
& System.Object^ FaceOrPlaneToSelect,  
& [Out] Component2^ InsertedComponent  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceOrPlaneToSelect
Plane or planar face
InsertedComponent
New part inserted as virtual component

Return Value

Error as defined by [swInsertNewPartErrorCode_e](#)

Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::InsertNewVirtualPart](#).

Example

[Insert New Virtual Component \(VB.NET\)](#)
[Insert New Virtual Component \(VBA\)](#)
[Insert New Virtual Component \(C#\)](#)
[Insert New Instance of Virtual Component \(VBA\)](#)
[Insert New Instance of Virtual Component \(VB.NET\)](#)
[Insert New Instance of Virtual Component \(C#\)](#)

See Also

[IAssemblyDoc Interface](#)
[IAssemblyDoc Members](#)
[IComponent2::IsVirtual Property](#)
[IModelDocExtension::IsVirtualComponent3 Method](#)
[IAssemblyDoc::InsertNewVirtualAssembly Method](#)
[IAssemblyDoc::InsertNewAssembly Method](#)
[IComponent2::MakeVirtual Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0